

Professional Summary

3rd-year B.Tech student in AI & ML with proven experience building and evaluating real AI systems. Currently integrating an LLM-powered diagnostic explanation layer into the Pneumonia Detection UI. Seeking an AI/ML junior role where practical problem-solving and fast execution matter.

Experience

•Smart Interviews

Jan 2026 – May 2026

Student Mentor

- Mentoring **500+** students in Data Structures & Algorithms across **6 hours/week** of structured doubt-resolution sessions.

Technical Skills

Languages: Python, C, C++, Javascript**ML/AI:** PyTorch, Scikit-Learn, Pandas, Numpy, PCA, Transfer Learning, Data Augmentation**Full -Stack:** React.js, Next.js, Node.js, Express.js, Fast API, WebSockets, HTML, CSS, OracleSQL, MongoDB**Tools & Platforms:** Git, GitHub, Google Colab, Kaggle, VS Code

Projects

•Pneumonia Detection – Hybrid ResNet-50 + Vision Transformer

Jan 2026 – May 2026

Deep Learning Medical Imaging System for Automated Pneumonia Detection

- Engineered a hybrid deep learning architecture combining ResNet-50 (CNN) with Vision Transformer (ViT) via transfer learning on the Kaggle Chest X-Ray dataset (**5,863 images**); achieved **96% accuracy, 97.8% F1, and 96.6% AUC-ROC** on the held-out test set — outperforming standalone ResNet-50 by **+14.1% accuracy**.
- Benchmarked against EfficientNet, VGG16, and DenseNet; the hybrid model attained the highest AUC-ROC across all architectures, with cross-validation on an additional Kaggle dataset confirming generalization.

•Network Monitoring Dashboard

Nov 2025

AI-Powered Network Monitoring & Anomaly Detection System

- Implemented a full-stack real-time network monitoring dashboard capturing live system metrics (bandwidth, latency, device connectivity) using **psutil** for real network data ingestion — not simulated — streamed to a React frontend via **WebSockets**.
- Designed and implemented a **Z-Score based anomaly detection engine** that statistically identifies abnormal traffic spikes and flags potential security events in real time, with visual alerts surfaced directly in the dashboard.

•Weather Forecasting Model

May 2025

Weather Forecasting & AI Insights System

- Achieved $R^2 = 0.998$ and RMSE of 0.7 on held-out weather data using a Random Forest Regressor pipeline — a **36% RMSE improvement** over the persistence baseline.
- Applied PCA to compress feature space from **13 to 6 components** while retaining 95% variance, reducing inference latency without meaningful accuracy loss.
- Integrated OpenWeatherMap API for live data ingestion, enabling real-time forecast generation on current conditions.

Positions of Responsibility

•Internet Society(ISOC), VNR VJIET

May 2025 – Present

Head of Events

- Organized **6 technical events** covering Web Technologies, GenAI/LLMs, and DevOps with **300+ attendees per event**. Led end-to-end logistics, speaker coordination, and student outreach.

Achievements & Certifications

•Diamond Coding Certification

2025

Smart Interviews

Certificate

- Ranked in the **top 5%** of 50,000+ coders, demonstrating strong proficiency in DSA, and problem-solving.

Education

•Bachelor of Technology in Computer Science and Engineering –AI&ML

2023 – 2027

VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad

CGPA: 7.84